ICP 05. Treatment and Refractory Disease

Companion reading handout for simultaneous listening.
Audio course on intrahepatic cholestasis of pregnancy. Episode five. Treatment and refractory disease.

Source Base

This episode is based on the Israeli ICP position paper, Gabbe's hepatic disease chapter, and the SMFM consult.

Opening Frame

Let us begin with a question that prevents bad counseling.

What exactly are we treating?

In ICP, treatment has several possible goals. We want to relieve maternal pruritus. We want to improve maternal biochemical cholestasis. We would like to reduce fetal bile-acid exposure if we can. And we want to prevent stillbirth.

Those goals sound similar, but they are not the same.

Ursodiol helps many mothers feel better. It often improves labs. But it is not proven to prevent stillbirth. Delivery timing remains the central fetal-risk intervention.

So the core treatment sentence is this.

Medication treats symptoms and biochemistry. Delivery ends exposure.

Ursodeoxycholic Acid

Now let us start with ursodeoxycholic acid, also called ursodiol, or UDCA.

UDCA is a hydrophilic bile acid. Its pharmacologic logic is elegant. Instead of leaving the mother with a more toxic cholestatic bile-acid pool, UDCA shifts the pool toward less toxic bile acids and promotes bile-acid transport and excretion.

Clinically, that means itch often improves and liver tests often improve. Gabbe states that ursodeoxycholic acid tends to normalize the bile-acid profile and decreases pruritus by stimulating bile-acid excretion. The Israeli position paper states that it reduces pruritus and improves laboratory results. SMFM recommends it as the first-line agent for maternal symptoms.

Evidence Nuance

Now the evidence nuance.

Earlier meta-analyses suggested possible improvement in some perinatal outcomes. But the large PITCHES randomized trial did not show a significant improvement in the composite outcome of perinatal death, preterm delivery before thirty-seven weeks, or neonatal unit admission for at least four hours. It did show greater improvement in maternal itch.

So do not tell a patient, this drug prevents stillbirth. Tell her, this is the best first-line medication for your itching and your liver tests, and we will still make the delivery plan based on bile-acid risk.

UDCA Dosing

Now dosing.

SMFM gives a starting dose of ten to fifteen milligrams per kilogram per day, divided into two or three daily doses. Common regimens include three hundred milligrams twice daily, three hundred milligrams three times daily, or five hundred milligrams twice daily.

The Israeli paper gives a practical starting dose of three hundred milligrams, three to four times daily.

If symptoms persist, SMFM allows titration up to a maximum of twenty-one milligrams per kilogram per day.

Now take the example that matters clinically.

Ursodiol three hundred milligrams four times a day is twelve hundred milligrams per day.

For a seventy kilogram patient, that is about seventeen milligrams per kilogram per day. It is within the SMFM maximum.

For a sixty kilogram patient, it is twenty milligrams per kilogram per day. Still within the maximum.

For a fifty kilogram patient, it is twenty-four milligrams per kilogram per day. That is above the usual ceiling.

So the answer is not, three hundred four times daily is always fine, and not, it is always too much. The answer is, check the weight, check tolerability, and understand the ceiling.

Why There Is a Ceiling

Why is there a ceiling?

Not because twenty-one milligrams per kilogram is a magic toxicity cliff. The ceiling reflects the studied and recommended therapeutic range. Above it, there is no proven fetal benefit, symptom improvement may plateau, and maternal side effects such as nausea and dizziness become more relevant. The dose ceiling is an evidence and tolerability boundary.

The expected timeline is also useful. Pruritus often improves within one to two weeks. Biochemical improvement usually takes three to four weeks. That delay matters. If the patient starts UDCA on Monday, do not expect Tuesday's bile acids to prove failure or success.

Refractory Symptoms

Now what if she remains miserable?

The first step in refractory ICP is not automatically adding a drug. The first step is to reassess.

Are we sure this is ICP? Is there a primary rash? Is there biliary obstruction? Is there hepatitis? Is there preeclampsia or HELLP? Is there acute fatty liver of pregnancy? Is there a medication injury? Are bile acids truly severe, or is the symptom burden severe with moderate bile acids?

Next, optimize UDCA up to the accepted dose ceiling if appropriate.

If disease remains refractory, the key adjunct to remember is rifampin, also called rifampicin.

Rifampin is familiar as an antibiotic, but in cholestatic disease its value is related to hepatic enzyme and transporter induction. Conceptually, it can increase pathways that help process and excrete bile acids. SMFM notes that rifampin can be combined with UDCA for refractory ICP with improvement in pruritus.

This is not a casual first-line medication. It is an adjunct for selected refractory cases, ideally with maternal-fetal medicine and hepatology involvement. Think about drug interactions and liver monitoring.

Cholestyramine

Now cholestyramine.

Cholestyramine has a beautifully simple mechanism. About ninety-five percent of bile acids are normally reabsorbed into the portal circulation. Cholestyramine binds bile acids in the gut and prevents reabsorption. In theory, that interrupts enterohepatic recycling.

But clinically, it is less attractive than the mechanism sounds. SMFM says it has limited impact on pruritus in ICP and a significant gastrointestinal side effect profile. The Israeli paper lists nausea, vomiting, abdominal pain, and bloating. It can also reduce vitamin K absorption.

That vitamin K issue is not a footnote. ICP itself can be associated with fat-soluble vitamin problems, and cholestyramine can worsen vitamin K deficiency. Near delivery, deficiency can increase bleeding risk. Gabbe specifically connects vitamin K deficiency from cholestasis or cholestyramine with postpartum hemorrhage risk and suggests monitoring prothrombin time when relevant and giving vitamin K as necessary.

Now S-adenosyl methionine.

S-adenosyl methionine has been used as an alternative or adjunct in cholestatic physiology. It may improve pruritus in some patients, but SMFM notes that it is less effective than UDCA. It is not the central first-line medication.

Antihistamines

Now antihistamines.

Hydroxyzine and diphenhydramine can help sleep. They may reduce the experience of nighttime itching through sedation. But they do not correct bile-acid transport. They do not lower fetal bile-acid exposure. They are comfort medications, not disease-modifying therapy.

Older and Supportive Treatments

Gabbe also lists phenobarbital among older treatments used to ameliorate pruritus or lower bile-acid concentration. In modern practice it is not the central resident-level answer. Think of it as a historical or uncommon adjunct, not as the first medication to reach for when a patient with ICP is itching.

Topical agents are similar. Menthol creams, calamine, cooling, and emollients may be soothing, but the itch is usually widespread and systemic. These treatments can support coping. They are not treatment of the cholestasis.

Plasmapheresis

Now plasmapheresis.

Plasmapheresis appears in the Israeli paper as an option in extreme cases. The concept is removal of circulating bile acids and other plasma solutes. But this is rare, resource-intensive rescue therapy. It is not routine ICP care.

Now we must return to delivery.

If a patient is thirty-five weeks with bile acids one hundred and forty and excruciating pruritus despite optimized medication, the answer may not be more and more medication. Depending on gestational age, local guidance, prior history, and fetal-maternal context, delivery may be the more rational treatment.

This is the mature way to think about severe ICP.

Escalate medication when ongoing pregnancy is appropriate. Do not escalate medication to avoid a delivery that is already indicated.

Final synthesis.

First, separate treatment goals. Itch relief, lab improvement, fetal exposure, and stillbirth prevention are not identical.

Second, UDCA is first-line. It improves pruritus and labs, but is not proven stillbirth prevention.

Third, dose UDCA by weight when possible. Start around ten to fifteen milligrams per kilogram per day, and do not exceed twenty-one milligrams per kilogram per day without a very deliberate specialist-level reason.

Fourth, refractory and adjunctive options include rifampin or rifampicin, cholestyramine, S-adenosyl methionine, antihistamines for sleep, older agents such as phenobarbital in uncommon settings, vitamin K when relevant, and rarely plasmapheresis.

Fifth, delivery is not just another treatment. Delivery removes the pregnancy driver and ends fetal exposure.

Medication treats symptoms and biochemistry. Delivery manages the fetal risk.

End of episode five.